Java 25 Features for Patterns

Season: 0 · Intro · Category: Language · Level: Beginner

Intent

- Use records, sealed types, pattern matching, and virtual threads as first-class pattern tools.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- New power tools that replace half the boilerplate older pattern books needed.

When to Use

- Any new pattern implementation on JDK 25+.
- The design force named in Intent shows up more than once.

When Not to Use

- Blindly enabling preview flags you do not need.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```
// see Used throughout src/
```

Modern Java 25 / Functional

- No preview features required in this repo — all on standard Java 25.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: Used throughout src/
```

Pitfalls

- Sealed + switch is often better than Visitor.
- Prefer virtual threads over hand-rolled pools for blocking I/O.

Interview Q&A

- Q: Records vs classes for DTOs? A: Records for transparent data carriers; classes when you need identity or rich mutable behavior.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Use records, sealed types, pattern matching, and virtual threads as first-class pattern tools.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: New power tools that replace half the boilerplate older pattern books needed. [0:40] Intent — State ...